Setup

Required Equipment

1. 8-channel audio system. The speaker layout is as specified below, but can be changed in config.scd in the “channelMaps” entry. Speakers are labeled 0 through 7.
```
FRONT
0 4 1
2   3
6 5 7
BACK
```
2. Webcam, tripod and projector with screen.
3. Launchpad X Controller.
4. Two performers.

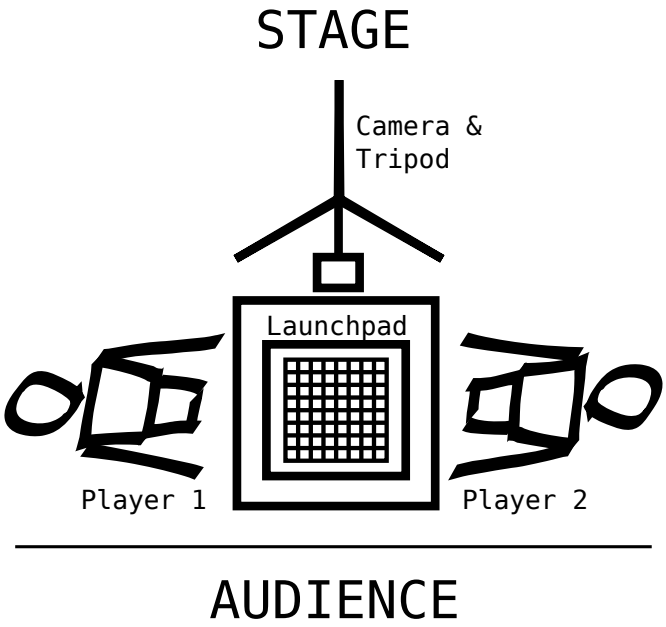
Installing the Software

This piece requires Max and SuperCollider. Max provides the visuals while SuperCollider provides synthesis.

1. Install SuperCollider version 3.14 or greater from https://supercollider.github.io/.
2. Install Max/MSP version 9 from https://cycling74.com/downloads.
3. Download the patch from https://github.com/tellmewyatt/horse/. To download, click the green “Code” button, then “Download Zip”. Extract the zip contents somewhere in your file system.

Stage Setup

The stage should be set up according to the diagram, with the launchpad in its upright orientation when viewed by the audience. The two performers will sit or stand across from each other. The camera should be placed directly above the Launchpad if possible, with the launchpad filling most of the vertical space in the camera’s video feed.



Software Setup

1. After plugging the Launchpad X in, set it to “Programming Mode” by holding the “Session Button” and then pressing the orange “Scene Launch” button. Tap the Session button again until the Launchpad is blank.
2. Open “live-visuals.maxpat” in Max. Click the “On/Off” toggle.
3. Open “horse.scd” in SuperCollider. Click anywhere in the code region and hit CMD+Enter. In a few moments you should see the Launchpad’s pads light up. You should also see video appear on the Max jit.world window. Tap pad 11 to verify sound.
4. Maximize the jit.world window. You are now ready to play the piece.

Pad Mapping

Each sound is a one-off and velocity controlled unless otherwise stated. A sound marked “sustained” requires the performer to hold the pad down until cutoff. A sound marked “triggered” will be sustained until tapped again. Triggered pads are marked purple. Switching between the first and second maps is accomplished by tapping on the pads marked “A” and “B” in the figure.

First Movement

- Column 1:
  - 81, 71, 61, 51: Vocoded “What am I?”
  - 41: Horse Gallops
  - 31: Horse Neighs
  - 21: “Snare” drum
  - 11: Bass drum
  - Column 2: Bright, flute-like sound
- Column 3: Vocoded horse neighs
- Column 4:
  - 84 and 74: Noisy fall
  - 64: Distorted Neighs (sustained)
  - 54: Distorted Gallops (sustained)
  - 44, 34, 24, and 14: Distorted “What am I?” (sustained)
  - Column 5: Distorted sine wave. The amount of distortion varies with the number of toggled pads in this column.

Second Movement

- Columns 1 – 3, 6 – 8: Bells (toggled)
- Columns 4 and 5: Noisy fall, enables a row
- Column 9: Stops column 5 from Map A and replaces it with a new vocal sample. After this, toggles this sound.

Performance Notes

For this explanation, a “row” will always refer to a group of pads who share the same first digit, and a “column” will refer to a group of pads who share the same second digit. This piece is composed of two movements. The first movement is for solo performer, while the second movement is for two. During the first movement, the first performer plays alone with the second performer offstage. The first performer will have pads 11, 21, 31, etc. closest to them, while the second performer will have pads 19, 29, 39, etc. closest to them when they walk on stage during the transition.

The piece roughly moves from column to column. The first performer starts at pad 81 and gradually introduces pads in the next three columns, and pads 14, 24, 34, and 44. When they have built enough energy they introduce pads 74 and 84. Finally, they press and hold pads 55 and 65. They wait until pad 65 hits a high volume point and then release it. They do this again for column 55, waiting until it reaches a high point, then releasing it, but this time they simultaneously play pad 11 and 15.

This initiates the transition between movements. The first player builds a line of toggled buttons in column 5, playing pad 11 each time. They then press the “B” map change button, switching the pad configuration to that of the second movement. At this point the second performer walks on stage, and looks at the first performer for some time before pressing 89 and 88 simultaneously. The two performers then take turns toggling the bells closest to them on, with the second performer toggling both columns 8 and 9.

Once this is done, the performers simultaneously press a pad in columns 4 and 5 to enable playing in that row. They gradually continue until all rows are unlocked. At this point, they can hit the pads in columns 1 through 3 and 6 through 8 with an open palm. The performers slow down and reduce energy. This should result in a random dispersion of toggled and untoggled bell pads.

The second player should wait for any accidentally cued sounds to end. Then, they press a pad in column 9 while simultaneously toggling on a previously untoggled bell pad in that column. The first player toggles on one of their bell pads in turn. In this way, the performers “fill in” any untoggled pads in rows 1 to 3 and 6 through 8 by toggling them on. If all the pads in the row are already toggled, the performer can untoggle and retoggle one. Once this row’s bell pads have all been toggled on, the performers move to another row and repeat this process. The piece ends with the performers toggling off the final column 9 pad and pads 14 and 15. The performers play on pads 11 and 18 slowly as the remaining sounds fade away.